

WEEK 3 PROGRESS UPDATE

Exploration of Speech Recognition, Multilingual Intent Datasets, and Initial Baseline Architecture

Research Progress Report

Prepared by	Y.Unmesha Kruthika	Date	21/08/2026
Reporting Period	Week 3	Status	Ongoing — Exploratory Phase

1. Objective

The objective of this week's work was to understand the different components required for developing an audio-to-intent understanding system, explore suitable existing models and datasets, and design an initial baseline architecture for future experimentation.

The main focus this week was on research, exploration, and understanding, rather than optimization or model development. The key components explored were:

- Whisper for speech recognition
- MASSIVE for multilingual intent classification data
- MuRIL for Indian-language text understanding and intent classification

Based on this exploration, an initial baseline architecture was proposed:

Audio → Whisper → Text → MuRIL → Intent

2. Understanding the Audio-to-Intent Problem

The overall objective of the project is to identify the intent behind a person's spoken input. For example, if a user says "Play a podcast", the system should understand that the user's intention is play_podcasts.

The input to the final system is therefore speech, while intent classification models generally operate on text. This led to the initial idea of separating the problem into two stages:

Speech
↓
Speech Recognition
↓

```
Text
↓
Language Understanding
↓
Intent
```

Therefore, two different capabilities are required: (1) understanding what was spoken, and (2) understanding what the speaker intends.

3. Exploration of Whisper

What is Whisper? Whisper is a multilingual automatic speech recognition model developed by OpenAI, introduced in the research paper "Robust Speech Recognition via Large-Scale Weak Supervision" by Radford et al., published at ICML 2023. Whisper was trained using approximately 680,000 hours of multilingual and multitask audio data.

The purpose of studying Whisper was to understand whether an existing multilingual speech-recognition model could be used as the first component of the proposed audio-to-intent pipeline.

4. How Whisper Converts Audio into Text

The basic processing performed by Whisper can be represented as:

```
Audio Waveform
↓
Audio Representation
↓
Log-Mel Spectrogram
↓
Transformer Encoder
↓
Encoded Speech Representation
↓
Transformer Decoder
↓
Text Tokens
↓
Transcription
```

4.1 Audio waveform. When a person speaks, the microphone captures variations in air pressure and converts them into a digital audio signal. The result is an audio waveform, representing how the signal changes over time. However, the raw waveform itself is not directly interpreted as words.

4.2 Log-Mel spectrogram. Whisper first converts the audio into a log-Mel spectrogram. A spectrogram represents the distribution of frequencies in the audio over time, and the Mel scale provides a frequency representation related to human auditory perception. The audio is therefore transformed from a time-domain

waveform into a time-frequency representation that contains useful information about the speech sounds present in the recording.

5. Whisper Encoder

The log-Mel spectrogram is provided to the Transformer encoder, which processes the acoustic information and creates an internal representation of the speech. It learns patterns related to:

- speech sounds
- phonetic characteristics
- pronunciation
- temporal relationships
- language-specific patterns
- contextual acoustic information

The encoder therefore converts a speech signal into a meaningful acoustic representation.

6. Whisper Decoder

The encoded speech representation is then provided to the Transformer decoder, which generates the transcription as a sequence of tokens. For example, for Hindi speech “एक पॉडकास्ट को प्ले करो”, the decoder generates tokens corresponding to the Hindi transcription, generating each token sequentially while using previously generated tokens and the encoded audio representation as context:

```
Encoded speech
↓
Decoder
↓
Token 1
↓
Token 2
↓
Token 3
↓
...
↓
Complete text
```

7. Multilingual Capability of Whisper

An important reason for exploring Whisper is its multilingual capability — the model can recognize speech in many different languages using the same general architecture. Whisper also performs language identification, allowing it to determine the language of the input speech.

For the proposed project, the important task is speech → text in the same language, rather than translating the speech into another language. This is important because the next component, MuRIL, needs to receive the original Indian-language text.

8. Whisper Model Variants

Whisper is available in multiple model sizes, including Tiny, Base, Small, Medium, and Large. The different versions mainly represent different model capacities and computational requirements — in general, increasing the model size provides greater modelling capacity but also increases computational requirements.

For the current project, the important observation at this stage is that Whisper provides multiple model configurations that can later be experimentally compared depending on available computational resources and transcription quality. At this stage, the focus was understanding Whisper and its role in the proposed architecture, rather than selecting a final Whisper configuration.

9. Exploration of MASSIVE Dataset

After understanding the speech-recognition component, the next requirement was to identify a suitable dataset for the intent classification component. The MASSIVE dataset was therefore explored. MASSIVE stands for Multilingual Amazon SLU Resource Package — a multilingual dataset designed for natural language understanding and spoken language understanding.

The original MASSIVE dataset contains approximately 1 million utterances, 51 languages, 60 intents, 18 domains, and 55 slot types, containing realistic user utterances corresponding to virtual-assistant-type tasks.

10. Intent Information in MASSIVE

The dataset contains multiple types of user intentions, for example:

```
play_music
play_podcasts
play_radio
alarm_set
weather_query
transport_taxi
calendar_query
music_query
email_query
cooking_recipe
```

For example, “Play some music” can correspond to `play_music`, while “Play a podcast” can correspond to `play_podcasts`. This makes MASSIVE useful for studying the intent-classification part of the proposed system.

11. Important Observation About MASSIVE

During exploration, an important distinction was identified: MASSIVE is primarily a text-based NLU dataset. It provides an utterance + intent label, rather than providing the complete audio recordings required for direct audio-to-intent training.

Therefore, MASSIVE can be used to investigate the text-to-intent component of the proposed system. It provides a useful starting point for building a baseline before attempting direct audio-based intent understanding.

12. Selection of Indian Languages

Since the project focuses on Indian languages, an initial subset of Indian languages was selected from MASSIVE. The five languages selected for the initial exploration were:

Language	MASSIVE Locale
Hindi	hi-IN
Telugu	te-IN
Kannada	kn-IN
Tamil	ta-IN
Malayalam	ml-IN

These languages were selected to create an initial multilingual Indian-language baseline. The intention at this stage is not to claim that these are the only relevant languages, but to begin experimentation with a manageable subset.

13. Selection of MuRIL

After exploring the dataset, the next question was: which language model should be used to perform intent classification on Indian-language text? For the initial baseline, MuRIL was selected.

MuRIL stands for Multilingual Representations for Indian Languages. It is a BERT-based multilingual language model specifically developed with Indian-language representation learning in mind, introduced in the research paper "MuRIL: Multilingual Representations for Indian Languages" by Khanuja et al.

14. Why MuRIL Was Considered Suitable

MuRIL was considered suitable for the initial experiment because:

- It is designed specifically for Indian languages.
- It supports multilingual representation learning.
- It can be fine-tuned for downstream NLP tasks.
- It is based on the Transformer architecture.
- It can be adapted to classification tasks by adding a classification layer.

Therefore, it provides an appropriate starting point for investigating Indian-language text → intent.

15. How MuRIL Works

MuRIL follows the BERT-style Transformer architecture. Its basic processing can be represented as:

```
Indian-language Text
↓
Tokenizer
↓
Token IDs
↓
Embeddings
↓
Transformer Encoder
↓
Contextual Representation
↓
Classification Layer
↓
Intent
```

16. MuRIL Tokenization

When an Indian-language sentence is given to MuRIL, the tokenizer first divides the sentence into tokens. For example, “एक पॉडकास्ट को प्ले करो” is converted into a sequence of tokens and then into numerical token IDs. These numerical representations can then be processed by the Transformer.

17. Transformer-Based Understanding

The token representations pass through multiple Transformer layers. The major mechanism used by the Transformer is self-attention, which allows the model to examine relationships between different parts of the sentence. For example, in “एक पॉडकास्ट को प्ले करो”, the model can learn relationships between concepts such as podcast + play, rather than treating every word independently. Through the Transformer layers, MuRIL generates contextual representations of the input sentence.

18. From Representation to Intent

MuRIL itself is a pretrained language representation model. To use it for intent classification, a classification layer can be added on top of the pretrained representation. In the proposed experiment, MASSIVE provides 60 intent classes, so the classification layer is designed to produce scores corresponding to the 60 intents.

Conceptually:

```

MuRIL
↓
Contextual Representation
↓
Classification Layer
↓
60 Intent Scores
↓
Highest Score
↓
Predicted Intent

```

19. Proposed Initial Baseline Architecture

Based on the exploration of Whisper, MASSIVE, and MuRIL, the first proposed baseline architecture for the project is:

```

AUDIO INPUT
↓
[ WHISPER – Speech Recognition ]
↓
Indian-Language Text
↓
[ MuRIL – Text Understanding ]
↓
Intent Classifier
↓
Intent Label

```

In short:

```

Audio → Whisper → Text → MuRIL → Intent

```

20. Reason for Choosing This as the First Baseline

This architecture was selected as the initial proposed baseline because it divides the problem into two understandable components.

Whisper handles: “What did the person say?”

MuRIL handles: “What does that sentence mean / what is the user's intent?”

This separation makes it possible to first understand and establish each component independently. It also provides a clear baseline against which later approaches can be compared.

21. Proposed Initial Experimental Direction

The immediate experimental direction is therefore to investigate this baseline:

```
Audio
↓
Whisper
↓
Indian-language transcription
↓
MuRIL
↓
60-class intent classification
```

The first objective is to understand how well this cascade can perform on the selected Indian languages and identify the behaviour and limitations of the baseline.

22. Current Status

The project is currently at the baseline exploration and architecture-design stage. The following components have been identified:

Component	Selected Approach
Speech input	Indian-language audio
Speech recognition	Whisper
Initial dataset	MASSIVE
Initial languages	Hindi, Telugu, Kannada, Tamil, Malayalam
Text understanding	MuRIL
Intent classification	MuRIL + classification layer
Initial architecture	Audio → Whisper → Text → MuRIL → Intent

23. Conclusion

This week's work focused on understanding the fundamental technologies and designing the first baseline for the project. Whisper was studied as the speech-recognition component responsible for converting spoken Indian-language input into text. MASSIVE was explored as a multilingual intent-classification dataset, and five Indian languages were selected for the initial experiment. MuRIL was then studied as the Indian-language text-understanding model for the intent-classification component.

Based on these explorations, the first proposed baseline architecture was established as:

Audio → Whisper → Indian-Language Text → MuRIL → Intent

The next stage of the project will involve implementing and experimentally evaluating this proposed baseline.

References

1. Radford et al., Robust Speech Recognition via Large-Scale Weak Supervision, ICML 2023 — Whisper.
2. Khanuja et al., MuRIL: Multilingual Representations for Indian Languages, 2021.
3. FitzGerald et al., MASSIVE: A 1M-Example Multilingual Natural Language Understanding Dataset with 51 Typologically-Diverse Languages, ACL 2023.